FlockML × CESC

Converting Power Utility Substation Networks into a Sovereign AI Compute Grid.

Executive Briefing & Infrastructure Proposal

Prepared for: Mr. Subir Verma (RPSG Group) · CESC House, Kolkata

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Why CESC? Four Direct Bottom-Line & Top-Line Returns

Solving immediate operational cost bottlenecks while creating a high-margin digital revenue stream.

1. DIRECT CLOUD SAVINGS

BOTTOM-LINE

Rs 3 - 5 Cr / yr

SMART METER TELEMETRY (RDSS)

3M+ smart meters stream readings every 15 mins. Processing locally on idle substation PCs eliminates recurring AWS GPU leasing.

2. PENALTY AVOIDANCE

GRID REGULATION

Rs 5 - 10 Cr / yr

DSM DEVIATION PENALTY REDUCTION

Real-time localized transformer load forecasting (650ms) predicts AC demand spikes, reducing State Load Despatch penalties.

3. ASSET PROTECTION

RELIABILITY

48h Advance Alert

TRANSFORMER BURNOUT PREVENTION

High-frequency thermal and harmonic monitoring detects winding degradation before physical transformer explosions.

4. NEW DIGITAL REVENUE

TOP-LINE

80%+ Gross Margin

SOVEREIGN AI COMPUTE CLOUD

Monetize excess substation power and dark fiber to sell private AI inference tokens to Kolkata banks, hospitals, and tech parks.

The Smart Meter Ingestion Challenge

National RDSS rollout generates 280 Million data points daily. Centralized cloud models create cost and regulatory traps.

THE CENTRALIZED CLOUD TRAP (AWS / AZURE)

- **Escalating Compute OpEx:** Running anomaly detection and phase balancing on AWS GPUs drains multi-crore budgets yearly.
- **CEA & CERT-In Compliance Exposure:** Critical power grid telemetry cannot legally reside on foreign multi-tenant public clouds.
- **Bandwidth & Latency Bottlenecks:** Egressing terabytes of raw meter data introduces lag during peak summer balancing.

THE FLOCKML SOVEREIGN EDGE SOLUTION

- **Zero Cloud Compute Bills:** Telemetry is processed locally at zonal substations (Chowringhee, Salt Lake, Howrah) on existing PCs.
- **100% Air-Gapped Grid Security:** Data stays strictly on CESC's private fiber network with zero foreign egress.
- **Sub-Second Localized Inference:** Neighborhood-level peak predictions execute in 650ms at the distribution transformer.

CESC’s Unfair Advantage: The Hidden Infrastructure Base

CESC already owns the three most expensive inputs required for artificial intelligence compute.

INPUT 01 · POWER

Rs 3.50

COST / KWH (GENERATION)

Commercial datacenters pay Rs 12-14/unit. CESC has generation-cost electricity. Power is 60%+ of AI datacenter operating expense.

INPUT 02 · SITES & FIBER

100+

SECURE SUBSTATION HUBS

Physical real estate across Kolkata and Howrah with established high-voltage feeds, security, and dark fiber right-of-way.

INPUT 03 · SILICON

5,000+

IDLE WORKSTATIONS & PCS

Substation SCADA terminals and office desktop machines sit completely idle from 7:00 PM to 7:00 AM every single night.

The Big Play: RPSG as India's Sovereign AI Champion

Leading the Rs 10,372 Crore IndiaAI Mission by pioneering Asia's first zero-CapEx utility compute grid.

THE NATIONAL SOVEREIGN MANDATE

- **The IndiaAI Opportunity:** The Government of India has committed **Rs 10,372 Crores** for domestic sovereign AI infrastructure independent of US cloud giants.
- **First-Mover Conglomerate:** While peers build capital-intensive imported GPU centers, RPSG can deploy the **First Sovereign Power-Grid AI Supercomputer** across domestic utilities.
- **Gateway to Central Deals:** RPSG's institutional standing unlocks national DISCOM deployments through MeitY and the Ministry of Power.

CONGLOMERATE MOAT

- **Permanent Energy Cost Moat:** Electricity arbitrage gives RPSG an unmatchable structural margin over traditional tech clouds.
- **Unregulated Digital Margins:** Transitions CESC from tariff-capped electricity distribution to high-margin Sovereign AI Cloud token sales.
- **Institutional Authority:** Executive relationships remove bureaucratic friction and establish RPSG as the national AI benchmark.

RPSG Group Synergies: Keeping AI Spend In-House

Powering artificial intelligence workloads across RPSG's multi-billion dollar conglomerate.

CESC POWER

Grid Telemetry

SMART GRID AI

Saves Rs 3-5 Cr/yr in cloud bills; prevents DSM penalties and transformer burnouts.

FIRSTSOURCE

BPO Support

$1B+ IT/BPO ARM

Runs customer support LLMs & voice transcription at ~70% lower cost than OpenAI.

WOODLANDS

Clinical AI

HEALTHCARE

100% private, on-premises radiology diagnostics with zero patient data leakage.

SPENCER'S

Retail Vision

150+ STORES

Real-time inventory computer vision and demand forecasting with zero server CapEx.

How FlockML Works: The Execution Engine

Zero-install WebGPU and WebAssembly runtime that executes across substation hardware.

SUBSTATION CLUSTER TOPOLOGY

```
[Node 01] Chowringhee HQ Terminal ──┐
[Node 02] Salt Lake Sec V Workstation ──┼─► FLOCKML ENGINE
[Node 03] Howrah Feeder RTU ──┘
```

- BitNet 1.58-bit Shaders: 80.2% VRAM reduction
- Work-Stealing Protocol: <5ms self-healing failover
- Browser Sandbox: Zero kernel drivers or OS changes

100% AIR-GAPPED CYBERSECURITY

- **Read-Only Shadow Ingestion:** Connects to read-only smart meter telemetry streams; never sends switching or breaker controls.
- **Zero Public Cloud Egress:** Model activations stay strictly on-soil within CESC's local network, fully compliant with CEA / CERT-In.
- **Zero CapEx:** Executes directly inside standard browser workers on existing Intel/AMD office workstations.

Verified Live Benchmarks & Economics

Performance metrics measured across a 3-node distributed desktop cluster.

SPEED

653 ms

TELEMETRY FORWARD PASS

Processes 50,000 smart meter streams and localized transformer load predictions in real time.

RESILIENCE

< 5.0 ms

WORK-STEALING FAILOVER

Zero dropped meter packets when a substation link drops. Remaining nodes steal workload instantly.

ECONOMICS

Rs 1.48 Cr

ANNUAL SAVINGS (PILOT BASE)

~72.4% compute cost reduction compared to equivalent AWS GPU instances.

The 14-Day Zero-Risk Substation Pilot

A non-intrusive evaluation on 10 idle office/substation PCs with zero interference to live SCADA.

DAYS 01 - 03

Phase 1: Setup

NODE PAIRING

Deploy FlockML worker on 10 idle PCs across Chowringhee, Salt Lake, and Howrah. Connect to read-only telemetry feed.

DAYS 04 - 10

Phase 2: Ingest

TELEMETRY BENCHMARK

Ingest 50,000 anonymized smart meter streams. Benchmark local peak load prediction against actual grid readings.

DAYS 11 - 14

Phase 3: Audit

EXECUTIVE REPORT

Deliver final Audit Report: Latency, cost comparison vs AWS, and 100% CEA cybersecurity compliance sign-off.

Next Step: 30-Minute Scoping Session

Kickstarting the 10-node pilot with the Chief Technology Officer or Head of Grid Operations.

THE IMMEDIATE ASK

1. Schedule a brief 30-minute technical scoping call with **CESC's CTO** or **Head of Grid Operations**.

2. Select 10 idle office/substation PCs across Kolkata.

3. Authorize the 14-day zero-risk read-only shadow telemetry pilot.

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